

R²-SAC: A Relaxation-and-Refinement SAC Agent for Stock Portfolio Trading

Xiaoyun Han

Financial Research Center

Hithink Flush Information Network Co., Ltd.

Hangzhou, China

Hanxiaoyun@myhexin.com

Jun Wang

Financial Research Center

Hithink Flush Information Network Co., Ltd.

Hangzhou, China

wangjun7@myhexin.com

Abstract—Deep reinforcement learning (DRL) has gained significant attention in stock portfolio management, in particular with the actor-critic framework that can generate continuous actions. This lends itself suitable for an end-to-end approach that produces a portfolio directly. However, there is no report that standard actor-critic agents can produce reliable excess profits in cross-market and cross-period tests. This paper introduces R²-SAC, a novel framework for stock portfolio trading that employs a ‘Relaxation and Refinement’ strategy to boost the famous Soft Actor-Critic (SAC) agent. *Relaxation* is employed in the stage of SAC policy generation, where the actor only produces coarse policy for individual stocks, and the real portfolio policy is formed by executing the coarse policy by a decision logic. *Refinement* is employed when executing the coarse policy in a real trading environment, where the decision logic takes into account turning points of the market and returns of individual stocks. Our experiments on four mainstream datasets demonstrated that R²-SAC significantly surpasses the baseline SAC in all the considered metrics, and harnesses significant excess profits in all the considered markets and test periods. Ablation studies showed that both relaxation and refinement are indispensable; they play significant roles by themselves and complement each other. The source code is available at <https://github.com/AI4QTrading/R2-SAC/>.

Index Terms—End-to-End Approach, Relaxation and Refinement, R²-SAC, Portfolio Management

I. INTRODUCTION

Portfolio management (PM) is the decision-making process that continually reallocates funds to a set of financial assets (including the cash account), with the goal of maximizing accumulated profits and minimizing potential risk. PM is more challenging than trading with single assets but may harvest excess profits by leveraging the correlation among assets. The portfolio theory was initially proposed by Markowitz in 1952 [1] and was further developed by other researchers [2], [3]. The essential of the theory is to estimate the expected return and risk of each asset and their correlation, and then assemble a high-profit and low-risk mixed asset accordingly. Two fundamental problems of this theory in practical usage are: (1) the price signal is highly noisy, making the estimation of the return and risk unreliable. (2) the market is dynamic, making the past estimation out-of-date when predicting the future.

Recently, a large body of intelligent trading agents based on Deep Reinforcement Learning (DRL) have been studied for

portfolio selection. The aim of these agents is set to harness as much profit as possible by interacting with the market. During these interactions, the agents are expected to learn and update their knowledge about the market, driven by the reward from the market.

Various DRL models have been proposed, among which the actor-critic framework has become increasingly popular such as A2C/A3C [4], DDPG [5], TD3 [6], [7] and SAC [8], [9]. This end-to-end learning arguably possesses a significant advantage compared to traditional methods that firstly predict the prices of assets and then make decisions according to the prices by a decision logic [10]. However, this advantage may be largely diminished when solving large-scale portfolio problems, especially when the market is not well developed or in transitional periods. This is because the reward signal is insufficient for the agents to learn both the potential returns of individual assets and the inter-asset competition for the limited capital. Therefore, a large amount of training samples are required to train a useful portfolio agent. Some methods could be used to alleviate the problem, e.g., the entropy regularization and the experience buffer replay have been demonstrated in the SAC agent to improve both sample efficiency and convergence speed [8], [9]. However, none of the present methods seems sufficient to train a workable portfolio agent, as demonstrated by recent empirical studies [11], [12] that show no clear evidence that portfolio agents can win the market consistently.

In this paper, we present a novel approach that can significantly improve the capacity of DRL portfolio agents. With the new approach, the actor-critic agent learns to solve a ‘relaxed’ trading problem rather than the original portfolio problem. Specifically, the agent focuses on learning how to trade individual assets without considering the capital restriction and produces a coarse policy. After that, a decision logic is designed to solve the portfolio problem by imitating human traders. Importantly, trading regulations and prior knowledge of the market and assets can be integrated into the decision logic to refine the policy. This relaxation & refinement approach allows the agent to focus on easier tasks, which may lead to faster and more stable training. We highlight that the actor-critic agent and the decision logic are integrated and trained as a whole system with the reward from the final policy.

II. R²-SAC FRAMEWORK

This section presents the overall framework of our proposed R²-SAC agent, which adopts a ‘Relaxation and Refinement’ strategy to boost SAC agents. Briefly, it involves a SAC agent with a relaxed policy network and outputs a coarse policy, and a refinement module to execute the coarse policy with additional market information extracted by a dual-auxiliary module. The architecture is illustrated in Figure 1, and the details are described below. Considering that our study is based on stock markets, we will specifically define an ‘asset’ as a stock or a cash account.

A. Relaxation module

The relaxation module is essentially a SAC agent, and the only difference is that the policy net outputs a coarse policy rather than a full portfolio. As usual, the trading task is defined as a Markov Decision Process (MDP), formulated as a quintuple (S, A, P, R, γ) . Here, S represents the set of environmental states; A denotes the set of actions that the agent may take; P is an unknown transition probability determined by the environment; R is the reward function, providing immediate feedback from the environment for an action taken in a given state. γ , ranging from 0 to 1, is the discount factor that represents the fading speed of a reward. More details are as follows.

State Space S : In our design, the state variable includes seven daily metrics for each stock: opening price, closing price, highest price, lowest price, trading volume, average trading amount, and turnover rate. Note that the state space of the agent involves a composition of the state variables of all the stocks in the market.

Action Space A : The action variable of a single stock comprises two elements, one for action and the other for quantity; both are continuous. The action element ξ_1 determines whether the stock should be ‘bought[0,1/3]’, ‘held[1/3,2/3]’, or ‘sold (2/3,1]’. The quantity element determines ξ_2 how much a stock should be bought or sold. The values range from 0 to 1 for each stock.

Reward Function R : The design of the reward function R in this paper is as follows. (1) Each episode involves a sequence of transactions and terminates if the net value of the whole assets is below zero, or the maximum number of transactions is reached. The reward is accumulated for each transaction. (2) After each transaction, a transaction reward is assigned based on the return rate calculated as the ratio of net value after and before the transaction step. (3) After each episode, an episode reward is augmented. If the episode fails (net value < 0), the episode reward is a large negative value of -10; otherwise, it is computed as the return rate of the capital after the entire episode.

Decision Logic and Action Policy: Based on the definition of the action space, the output of the policy net (actor) does not form a regular portfolio, but a coarse policy for each stock. A decision logic is designed to form an action policy for portfolio selection, by executing the coarse policy shown as follows: (1) For each stock, if the action element ξ_1 in the action

variable indicates buying, then allocate $\xi_2\%$ of the amount in the cash account to buy the stock; (2) For each stock, if the action variable indicates selling, then sell $\xi_2\%$ shares of the stock; (3) Calculate the net value of the account. The episode terminates if the net value is negative or the maximum number of transactions is reached; otherwise, go to the next transaction.

With the above definition, the SAC agent can be trained following the process presented by [8]. It should be noted that the relaxation & refinement approach can be combined with any DRL training algorithm, though we focus on the SAC algorithm in this study.

B. Dual-auxiliary module

The Dual-auxiliary module comprises two components: the market turning point prediction component and the stock return scoring component. The former predicts the market turning days based on the macro market information, while the latter forecasts the rise or fall for individual stocks based on micro stock information.

1) Market turning point prediction: Considering the noisy and non-stationary nature of markets, a turning point prediction model is designed to identify if the trend of the market will change in the next day. An accurate prediction for the turning point offers potential extra profits.

The prediction model is based on Temporal Convolutional Networks (TCN) and with the self-attention mechanism. The TCN [13] employs temporal convolutions to extract salient features related to the underlying dynamics of the market. To counteract overfitting, a dropout layer is integrated into the TCN. The self-attention mechanism [14] is applied to model long-term dependency. Finally, a fully connected layer is used to predict whether the next trading day is a turning point.

2) Stock return scoring: The stock return scoring involves two steps. The first step leverages a multi-dimensional Hawkes process [16] to analyze inter-stock correlations and construct a correlation graph among stocks. The second step uses a Graph Attention Network (GAT) to propagate the information among the stocks according to the correlation graph constructed by the Hawkes process. GAT outputs the potential return of every stock on the next trading day.

The Hawkes process is a point process used to describe a series of time-evolving events, where each event contributes to the likelihood of the appearance of the subsequent events, where the contribution diminishes as time progresses. This model is then suitable for modelling stock prices if we regard a significant positive or negative return as an event. To model the events of multiple stocks, a mutually exciting multi-dimensional Hawkes process can be used [17], where each dimension is a Hawkes process, and events in one process impact events in others. Mathematical details of the Hawkes process and the multi-dimensional version are beyond the scope of the paper; readers can refer to [16], [17].

Applying the multi-dimensional Hawkes process to model the inter-stock correlation is straightforward. Denote the stock

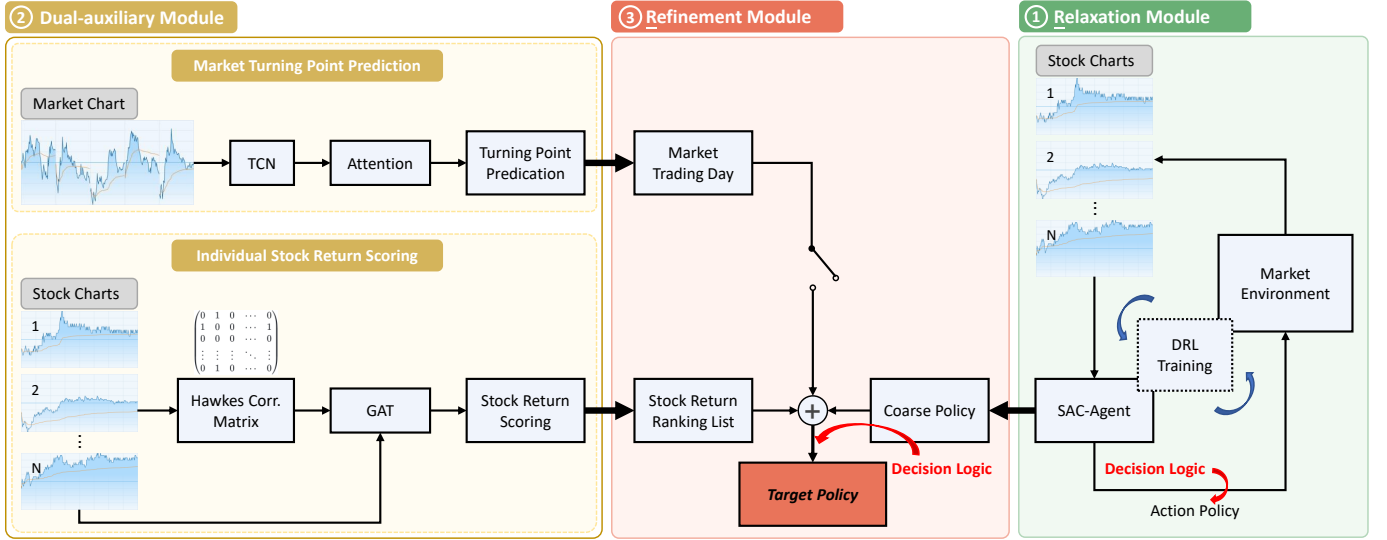


Fig. 1: The framework of our proposed R²-SAC agent. It consists of three modules: (1) Policy Relaxation Module (left); (2) Dual-auxiliary Module (right); and (3) Policy Refinement Module (middle).

price at time t by $v_t = (v_t^{(1)}, v_t^{(2)}, \dots, v_t^{(U)})$, where U is the total number of stocks in the market. For the u -th stock, an event is defined when the duration of $|v_t^{(u)}| \geq \delta$ exceeds ε , where δ and ε are two hyperparameters. By identifying all the events in all the stocks and sorting them chronically, the inter-stock correlation can be computed as the mutual contribution of the events from any two assets. This leads to a graph where each vertex represents a stock, and the edge weight is set to 1 if two stocks represented by the vertexes of the edge are correlated according to the Hawkes process; otherwise, it is set to 0.

After obtaining the correlation graph, a GAT model is used to predict the return rate of each stock. This model utilizes the Graph Attention Layer (GAL) to aggregate information of different stocks [18].

C. Refinement module

The Refinement module forms a target policy based on the coarse policy generated by the relaxed SAC agent and the auxiliary information produced by the dual-auxiliary module. The specific process is as follows:

(1) The market turning point prediction component in the dual-auxiliary module is used to predict whether the current day is a ‘turning day’. If it is not, no trading is performed.

(2) If the day is deemed a trading day, the stock returns produced by the dual-auxiliary module are used to rank the stocks, from the highest to the lowest.

(3) Finally, the decision logic described in the relaxation module is employed to produce true portfolio actions. The decision process will execute the buy and sell coarse actions produced by the SAC agent, following the rank list produced in (2).

III. EXPERIMENT

In this section, we will validate the performance of the proposed R²-SAC framework by conducting extensive experiments on four stock market datasets with several evaluation metrics. The consistent achievement on excess return rate demonstrated that the new method is reliable and effective.

A. Data and setting

1) *Data*: Our experiments are conducted on four stock market datasets: *CSI 500 index*, *CSI 300 index* in the Chinese A-share market, the date in the dataset is 2015.01-2023.06, *NASDAQ 100 index*, *DJIA 30* in the U.S. market, the date in the dataset is 2017.01-2023.08. Using rolling training and cross validation methods, the data from the last 3 years will be used for testing, while the remaining data will be used for training and validating the model. It is worth mentioning that for all four datasets, the test period spans from mid-2020 to mid-2023, a period full of accidents and chaos such as the COVID-19 pandemic and the Ukraine war. To make the model up to date, we split the test period into 3 sub-periods, and for each sub-period, a specific model is trained using data from about 5 years before. Moreover, we refrained from online updating the model in the test period, to test the generalizability of the system.

2) *Systems*: We developed six SAC-based DRL systems to validate the proposed R²-SAC framework, as presented below.

SAC: The vanilla SAC algorithm.

SAC(T): The dual-auxiliary module is enabled to improve the policy. As SAC is end-to-end, only the market turning point prediction can be used, i.e. the trading is only performed on market turning days.

Relax-SAC: A relaxed SAC agent proposed in the Relaxation module, which outputs a coarse policy and the decision logic is employed to produce a valid policy.

TABLE I: Performance comparison on four markets with different systems.

	CSI 500								CSI 300							
System	CW $\uparrow$	APY $\uparrow$	CER $\uparrow$	MD $\downarrow$	AVO $\downarrow$	ASR $\uparrow$	CR $\uparrow$	SoR $\uparrow$	CW $\uparrow$	APY $\uparrow$	CER $\uparrow$	MD $\downarrow$	AVO $\downarrow$	ASR $\uparrow$	CR $\uparrow$	SoR $\uparrow$
Market	1.04	0.01	0.00	0.32	0.19	0.16	0.01	0.21	0.95	-0.17	0.00	0.40	0.20	0.01	0.00	0.02
SAC	1.25	0.11	0.12	0.36	0.21	0.57	0.06	0.81	1.31	0.16	0.30	0.35	0.24	0.70	0.09	0.89
SAC(T)	1.70	0.24	0.56	0.31	0.19	1.04	0.12	1.58	1.82	0.29	0.81	0.36	0.28	0.89	0.12	1.35
Relax-SAC	1.50	0.15	0.36	0.36	0.21	0.70	0.07	1.11	1.58	0.20	0.65	0.29	0.22	0.74	0.10	1.05
Relax-SAC(T)	2.54	0.47	1.34	0.21	0.20	1.41	0.21	2.30	2.16	0.37	1.26	0.23	0.22	1.10	0.15	1.70
Relax-SAC(S)	2.39	0.44	1.15	0.20	0.22	1.12	0.19	1.84	1.89	0.31	1.02	0.25	0.21	0.98	0.14	1.48
R ² -SAC	3.24	0.67	2.00	0.16	0.19	1.80	0.26	3.01	2.39	0.45	1.44	0.20	0.21	1.31	0.20	1.94
	NASDAQ 100								DJIA 30							
System	CW $\uparrow$	APY $\uparrow$	CER $\uparrow$	MD $\downarrow$	AVO $\downarrow$	ASR $\uparrow$	CR $\uparrow$	SoR $\uparrow$	CW $\uparrow$	APY $\uparrow$	CER $\uparrow$	MD $\downarrow$	AVO $\downarrow$	ASR $\uparrow$	CR $\uparrow$	SoR $\uparrow$
Market	1.32	0.11	0.00	0.36	0.25	0.50	0.04	0.73	1.25	0.11	0.00	0.22	0.18	0.54	0.04	0.78
SAC	1.51	0.20	0.13	0.36	0.23	0.77	0.09	1.06	1.20	0.10	-0.07	0.19	0.14	0.80	0.09	1.11
SAC(T)	1.78	0.30	0.41	0.18	0.21	1.16	0.18	1.62	1.58	0.21	0.34	0.18	0.16	1.07	0.16	1.80
Relax-SAC	1.57	0.20	0.19	0.24	0.21	1.06	0.15	1.47	1.52	0.18	0.26	0.23	0.22	0.91	0.12	1.61
Relax-SAC(T)	1.86	0.30	0.48	0.18	0.20	1.22	0.18	1.77	1.63	0.24	0.65	0.20	0.19	1.00	0.13	1.71
Relax-SAC(S)	1.71	0.28	0.34	0.23	0.25	1.10	0.16	1.46	1.53	0.19	0.27	0.22	0.21	0.92	0.13	1.64
R ² -SAC	2.09	0.37	0.72	0.10	0.17	1.77	0.27	2.68	1.78	0.32	0.72	0.18	0.18	1.21	0.16	1.91

Relax-SAC(T): Relax-SAC augmented by the market turning point prediction to select the trading days in the decision logic.

Relax-SAC(S): Relax-SAC augmented with the stock return scoring component to rank the stocks when executing the coarse policy in the decision logic.

R²-SAC: Our proposed relaxation & refinement SAC, with both the market turning point prediction and stock return scoring used to improve Relax-SAC. It is essentially a short name for Relax-SAC(TS).

3) *Metrics:* Following previous studies from [20]–[23], seven metrics are selected to evaluate the performance of PM. Cumulative Wealth (CW), Cumulative Excess Return (CER), and Annualized Percentage Yield (APY) are used to measure the portfolios' returns. Maximum Drawdown (MD) and Annualized Volatility (AVO) are used to measure the portfolios' risks. Annualized Sharpe Ratio (ASR), Calmar Ratio (CR), and Sortino Ratio (SoR) are used to measure the risk-adjusted returns. Among these, higher values of CW, CER, APY, ASR, CR, and SoR indicate better performance, and lower values of MD and AVO signify superior performance.

B. Main results

The performance of four markets with different systems in terms of seven metrics is reported in Table I, and the temporal evolution of the accumulative returns is reported in Fig. 2. Generally, almost all the SAC systems outperform the market (the Blue line). This indicates that SAC-based portfolio agents are reliable in most cases. However, the standard SAC is risky and may be worse than the market in some circumstances, as is the case with DJIA30. And the proposed R²-SAC consistently exhibits superior performance across nearly all metrics in all datasets, particularly demonstrating a notable stable alpha relative to the market and smaller drawdowns.

IV. CONCLUSIONS

We presented a novel Relaxation-and-Refinement framework for portfolio management. The new framework tackles the portfolio problem in two phases: a relaxation phase that

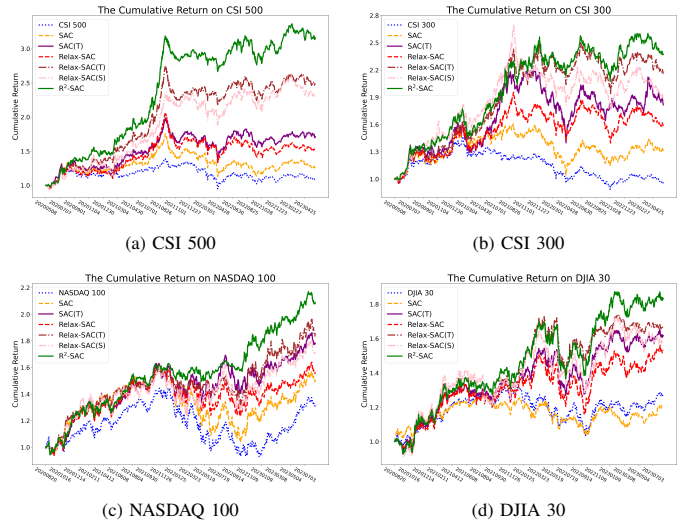


Fig. 2: The cumulative returns on four markets.

generates coarse policy, and a refinement phase that turns the coarse policy into a portfolio policy by a decision logic that may involve multiple information and knowledge. The framework was used to construct a Soft Actor-Critic (SAC) agent, which we named R²-SAC. Experimental results on four major stock market datasets demonstrated that R²-SAC significantly outperformed both the market index and the SAC baseline in all the markets and all the trading periods. In particular, we found that the market turning point information is highly effective and can be well integrated into R²-SAC by the decision logic. In the future, we plan to apply the Relaxation-and-Refinement framework to other deep reinforcement learning algorithms to verify its generality. Moreover, other auxiliary information will be used to improve portfolio selection by using the new framework.

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